UE23CS320B – Capstone Project Phase – 2

SEMESTER - VI
END SEMESTER ASSESSMENT

Project Title : A Unified Framework for Deep Multimodal Audio-Visual Steganography
Project ID : PW26_SBN_07
Project Guide : Dr. Surabhi Narayan
Project Team :

- Karthik Prabhu
- Harish Srinivasan
- Kappagantula Sai Manogna
- K S Arulmozhi Varma

Outline

- Abstract and scope.
- Suggestions from phase-2 review 2
- Implementation details (Technical)
- Results and demo
- Project Progress Plan for Phase 3 and Phase 4
- References

Abstract and Scope

- **PROBLEM STATEMENT -** Design a deeplearning-based **multimodal** steganography framework that can **jointly embed** and **extract** a high-capacity secret payload (especially a full video, including imagery and audio) within a host video by hiding information **across both the video's frames and its audio track**, while maintaining imperceptibility in each modality and robustness against common distortions (e.g. compression).

- **SCOPE** :
  - Implement Audio-INN + Video-INN as modular components
  - Per-chunk encode/decode loop (no drift, consistent k indexing)
  - Time-synchronous chunking
  - Process the clip in fixed sized chunks
  - Training + evaluation:
    - Stego quality (audio/video similarity vs cover)
    - Secret recovery quality (audio/video similarity vs secret)
    - Runtime feasibility on a single GPU setup

Suggestions from Phase 2- review 2

Panel Suggestions

- Explore multiple steganography approaches before finalizing the methodology
- Perform comparative analysis across different domains
- Provide justification for selecting the final approach based on outcomes

Work Done Based on Feedback

- Video Steganography: Explored a CNN-based architecture
- Audio Steganography: Explored frequency-domain spectrogram hiding method

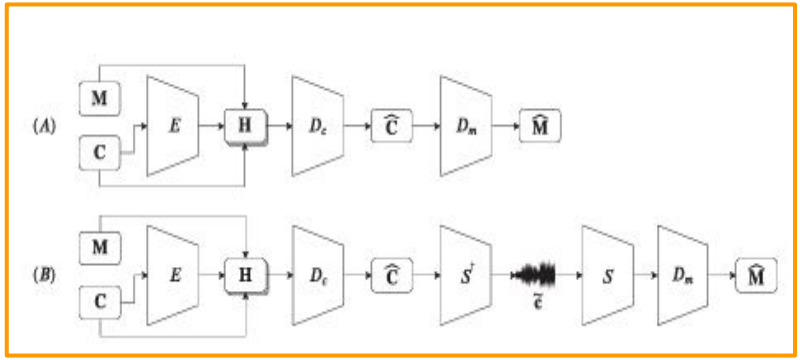
Video - Work Done Based on Feedback

Alternate Methodology : CNN Autoencoder
1. Image-based Training
Model trained on image pairs (cover + secret)
Uses resized images (e.g., 224×224)
2. Encoder Network
Combines cover image + secret image
Produces stego image
3. Decoder Network
Takes stego image as input
Recovers approximate secret image
4. Video Processing (Frame-wise)
Video is split into individual frames
Model is applied frame-by-frame
Processed frames are recombined into video
6. Output Pipeline
Stego frames → reconstructed into video

Why Our Video Methodology is Superior
1. Reversibility vs Approximation
Our Method: INNs → mathematically reversible, near-lossless recovery
Repo: CNN autoencoders → approximate, lossy recovery
2. Temporal Awareness vs Frame-wise Processing
Our Method: Sliding window → uses past + present + future frames
Repo: Frame-by-frame → no temporal context
3. Feature Fusion vs Naive Concatenation
Our Method: Wavelet / learned transforms → better feature mixing, fewer artifacts
Repo: Simple concatenation → weaker embedding quality
4. Deterministic vs Unstructured Pipeline
Our Method: Fixed, consistent encoding/decoding → reliable reconstruction
Repo: Frame extraction pipeline → no strict alignment guarantees

AUDIO - Work Done Based on Feedback

- Explored a deep learning-based audio steganography approach inspired by Hide & Speak - audio steganography purely in the frequency domain
- Proposes a deep speech steganography pipeline with Encoder (E) + Carrier Decoder (Dc) + Message Decoder (Dm) to hide a secret speech message inside a carrier speech and then recover it.
- Core technical aspect : inserts differentiable STFT + inverse-STFT (iSTFT) into the training loop, so the model learns hiding that remains decodable after spectrogram→waveform reconstruction (addresses the STFT-phase/overlap mismatch issue).
- Observations:
  - Reconstruction was lossy with noticeable degradation
  - Model lacked guaranteed invertibility, affecting accuracy
  - Performed inferior to INN-based approach (prototype) in maintaining quality and capacity





PROPOSED METHODOLOGY

- Video-in-Video Steganography INN
- Audio-in-Audio Steganography INN
- Temporal Synchronization Module

Video-in-Video Steganography INN

Goal: Hide one secret video inside a cover video using an Invertible Neural Network (INN) so decoding can near-exactly recover both.
Windowing (temporal context): Use a sliding window of length L around time t. The encoder processes the cover+secret window but outputs only the center stego frame **stego[t]**.
Feature Fusion (to reduce concat artifacts): A swappable TransformFusion module:

- **Fixed option**: Haar wavelet + learnable 1×1 conv mixing
- **Learnable option**: lifting / pseudo-wavelet via strided convolutions

Invertible Core (two-stream coupling): Operates on Cover features C and Secret features S with reversible coupling blocks:

- **Additive coupling**: C_{k+1} = C_k + g_s(S_k)
  Affine coupling: S_{k+1} = S_k * exp(clamp(g_scale(C_{k+1}))) + g_shift(C_{k+1})
- Uses lightweight subnets (e.g., depthwise separable / small 3D convs) + invertible 1×1 permutations between blocks.

Redundancy Estimator (key trick for single-frame output):

- In real invertible hiding, the stego frame may not be enough by itself to perfectly reverse everything — so the method uses an extra "helper" latent called Z (redundancy).
- Instead of sampling , predict missing redundancy $\hat{z}$ from stego context using a small residual CNN .
- Feed $\hat{z}$ into the reverse pass to reconstruct the secret.

Decoder "broadcast trick": Since decoder expects a window-shaped input, repeat stego[t] into a pseudo-group, can be modified so each copy is distinguishable.
Training objective (single-secret):

- **Forward loss**: keep stego close to cover (L1 + (1 − MS-SSIM))
- **Backward loss**: reconstruct secret (and optionally cover) (L1 terms)

What you get:

- Encoder: cover + secret → stego (visually similar to cover)
- Decoder: stego → recovered secret (+ recovered cover) via INN reverse + predicted $\hat{z}$

➢ Audio-in-Audio Steganography INN

1) Audio Preprocessing

- Cover and secret audio resampled to **16 kHz**
- Segmented into **1-second chunks** (16,000 samples)
- Scrambling within a secret chunk (simple encryption)
- Normalized and paired as network input

2) Invertible Neural Network (INN)

- Input: *(cover chunk x1, secret chunk x2)*
- Processed through a sequence of **invertible blocks**
- Each block uses:
  - **Coupling layers** (split, scale(s) , shift(t))
  - **Affine transformations**
  - **Channel shuffling** for full mixing

$$y_2 = x_2 \odot \exp(s) + t$$

$$x_2 = (y_2 - t) \odot \exp(-s)$$

3) Output Generation

- **Stego audio (y1)** : perceptually similar to cover, contains hidden secret
- **Auxiliary latent variable (y2)** : regularized to follow Gaussian distribution $\mathcal{N}(0, I)$ **;** Stores excess information for invertibility

4) Time–Frequency Loss (Applied during forward pass) : The model calculates the difference between the original cover and the generated stego audio in both the time domain (raw samples) and the frequency domain (using STFT). This ensures that the embedding process doesn't just minimize mathematical error, but also minimizes audible artifacts that the human ear might detect.

Temporal Synchronization Module

- Audio and video steganography are performed by *independent frameworks*, which can cause **audio–video drift** if temporal correspondence is not explicitly enforced
- We chunk both audio and video into fixed 1-second segments. Audio chunks consist of exactly 16,000 samples at 16 kHz, while video frames are grouped by timestamp using floor(time). All embedding and extraction is performed independently per chunk, ensuring strict temporal alignment.

Chunk Index Assignment

- Each 1-second interval is assigned a **global chunk index *i***, such that : `VideoChunk_i ↔ AudioChunk_i`
- This index represents the *same real-time interval* in both streams ensuring **one-to-one temporal correspondence**

For each index *i*:

- SecretVideoChunk_i is embedded into CoverVideoChunk_i
- SecretAudioChunk_i is embedded into CoverAudioChunk_i

Then, Concatenate all stego frames and Concatenate all stego audio chunks in time order

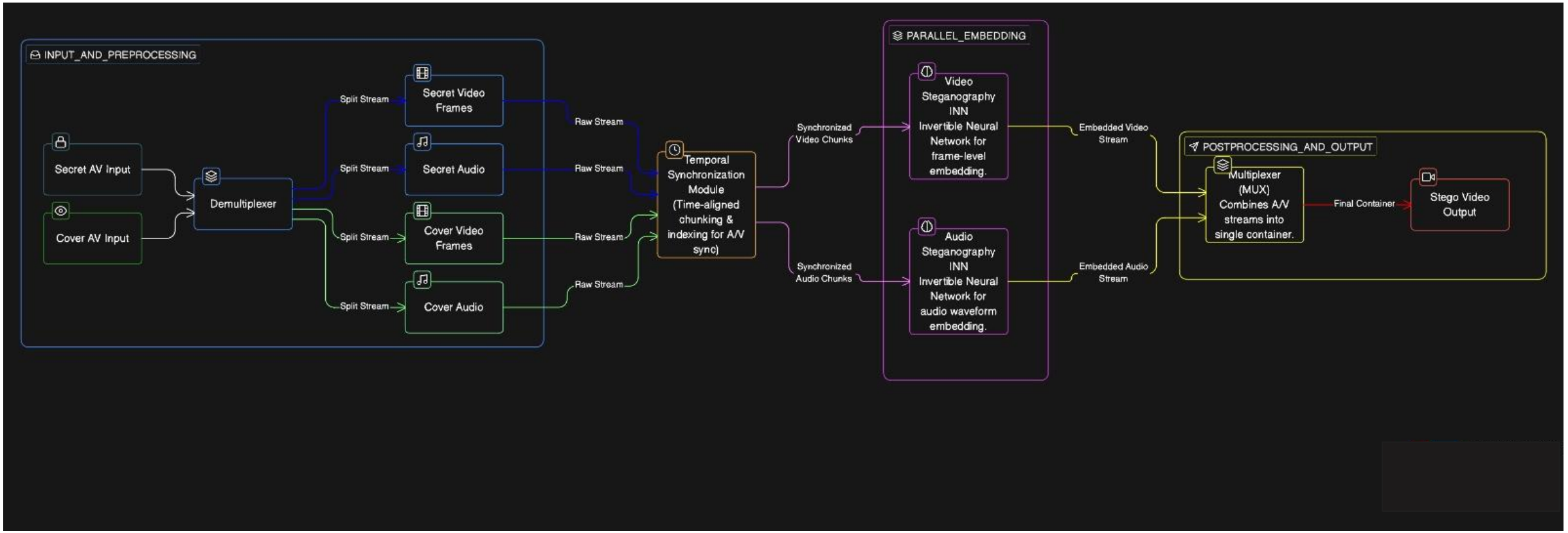
Decoding & Reconstruction

- During extraction, the same chunk indexing is applied, allowing recovered audio and video chunks with the same index to be recombined without drift even incase of variable frame rates.

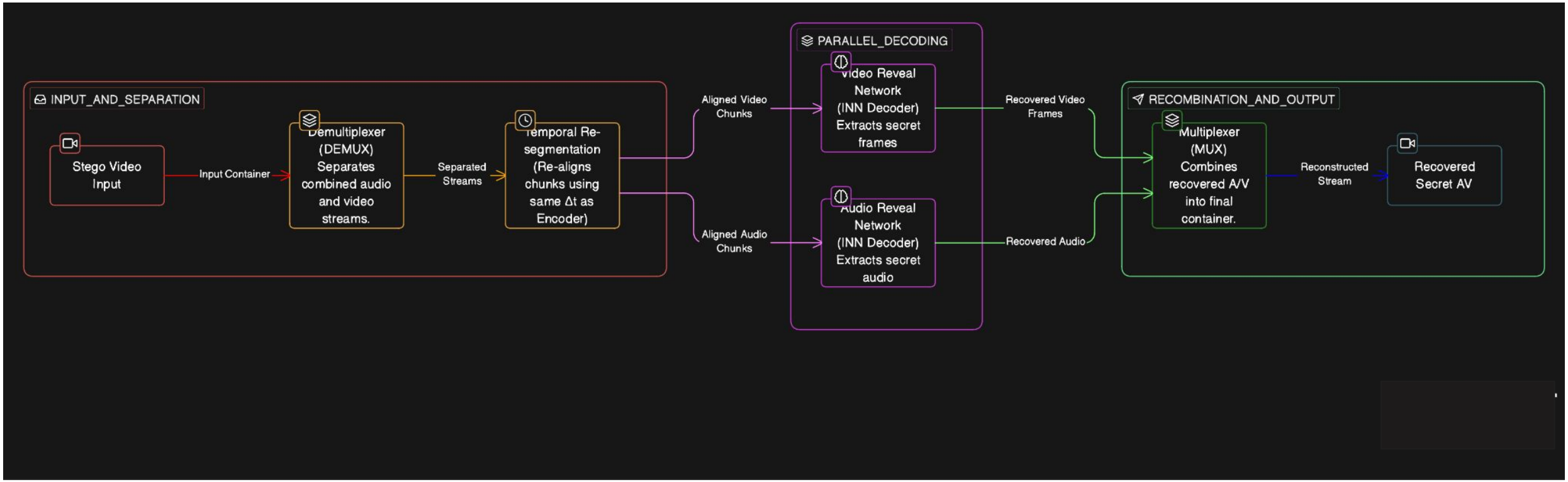
Why is this synchronization required?

- Say, first 200 ms of stego video removed, can I still decode aligned chunks? YES because with chunk indexing implemented decoding starts from next complete 1-second chunk
- Stego video is re-encoded from **30 FPS → 29.97 FPS** can I still keep alignment? YES since chunking is performed using time-based segmentation.

ENCODER ARCHITECTURE



DECODER ARCHITECTURE



Results and demo

```
### Test run summary

- Latest test summary file: `code/
results/train_LF-VSN_1video_proto/
metrics_test_summary.json`
- Checkpoint used in that test: `.../
models/19000_G.pth`
- Test sample count: `200`
- Test metrics:
  - PSNR Cover: `22.6197`
  - PSNR Secret: `22.5737`
  - PSNR Stego: `22.5418`
  - SSIM Cover: `0.7103`
  - SSIM Secret: `0.7069`
  - SSIM Stego: `0.6857`
```

```
Held-out eval showed approximately:

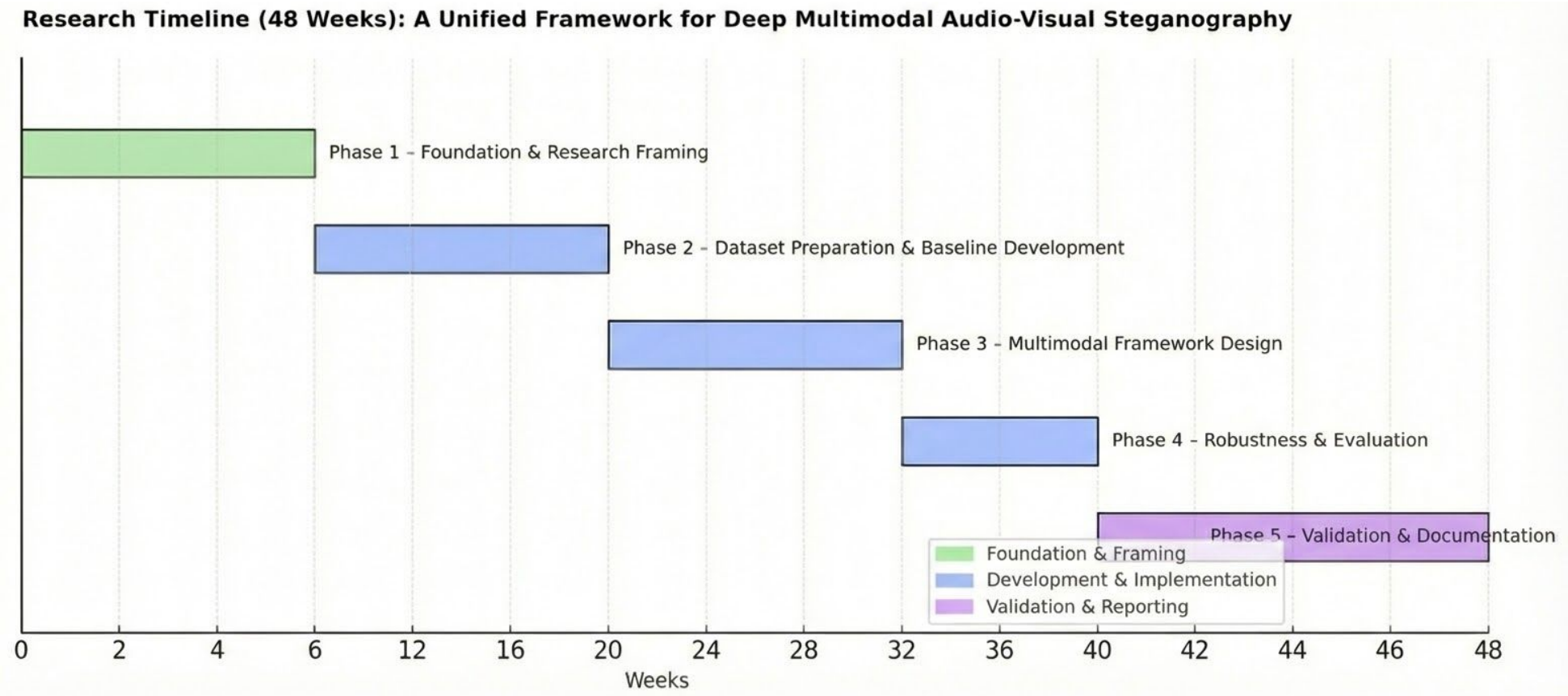
- float32: stored secret SNR 137 dB
- float16: stored secret SNR 74 dB
- int8: stored secret SNR 43 dB

So all three work well with stored r.
```

Individual contribution

NAME	CONTRIBUTION
KARTHIK PRABHU	Implementing Video-INN encoder with temporal windowing and feature fusion ensuring stego generation with minimal visual distortion
ARULMOZHI VARMA	Implementing INN-based decoding with redundancy estimation ensuring accurate secret recovery and improved reconstruction quality
HARISH SRINIVASAN	Designing time-based chunking and global indexing ensuring robust audio-video alignment and prevention of temporal drift
KAPPAGANTULA SAI MANOGNA	Implementing Audio-INN encoding and decoding with invertible coupling blocks ensuring high-fidelity embedding and accurate audio reconstruction

Project Progress Plan for Phase 3 and Phase 4



References

- Mou, C., Xu, Y., Song, J., Zhao, C., Ghanem, B., & Zhang, J. (2023, April 24). Large-capacity and flexible video steganography via invertible neural network. arXiv.org. https://arxiv.org/abs/2304.12300

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- WAVINWAV: Time-Domain speech hiding via invertible neural network. (n.d.). https://arxiv.org/html/2510.02915v1



THANK YOU